CASE STUDY: INTELLIGENT SYSTEMS IN PRODUCTION

REVISED TOPIC

AOI Decision Gate

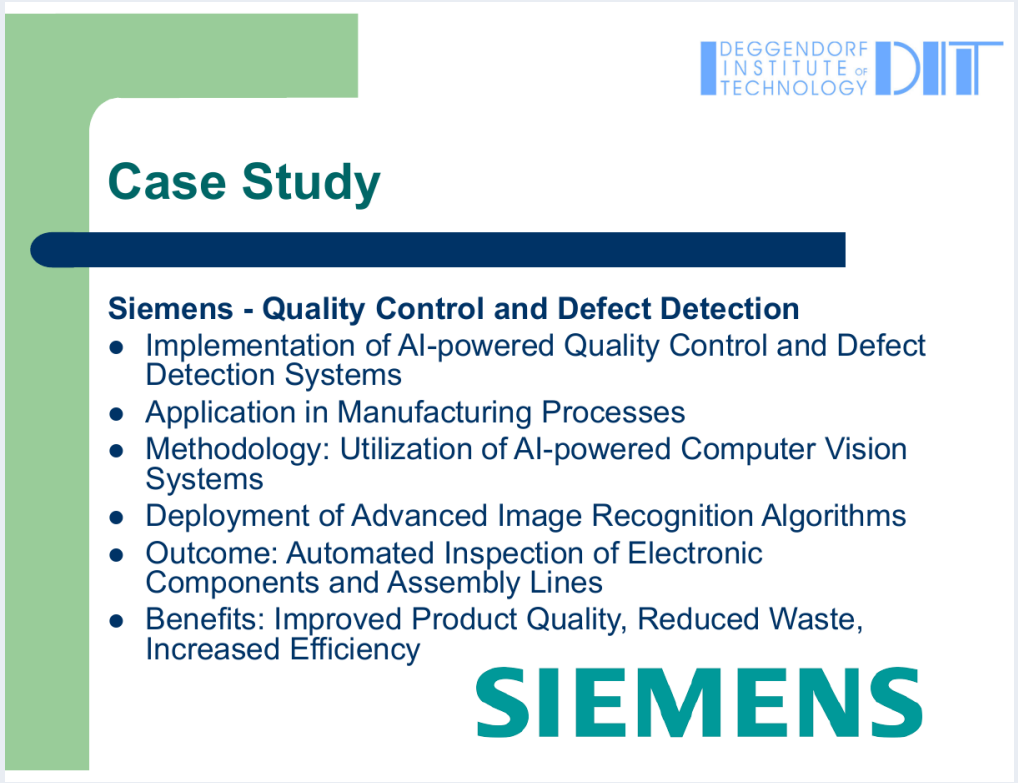
AI decision gate for false-call reduction in automated optical inspection of electronics assembly

Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas

Team 1 · Guided by Dr. Hamidreza Heidari · 23 October 2026 · github.com/bodhi584/aoi-decision-gate

WHERE THE TOPIC COMES FROM

Based on the lecture example: Siemens, p. 18



Lecture slide	Our project
AI-powered quality control and defect detection	**AI decides: real defect or false call**
Application in manufacturing processes	**SMT electronics assembly, after soldering**
AI-powered computer vision systems	**Works on the AOI camera system's measurements**
Advanced image recognition algorithms	**Classifier on AOI image measurements**
Automated inspection of electronic components	**Real Siemens line data (Bavaria)**
Improved quality, reduced waste, efficiency	**Fewer manual checks, no extra defects released**

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THE INDUSTRIAL PROBLEM

Too many false alarms after optical inspection, today

98 %

of the boards flagged by the AOI machine are fine (false calls) in the Siemens line data.

Every one goes to a manual inspection station: “a major manual labour effort” (Pfab et al. 2024, Siemens).

21 Oct 2026 · Siemens and NVIDIA, Erlangen electronics factory

AI second check of AOI results: is it a real defect or a false call?

Project targets: about 40 % fewer false calls, reduced manual verification.

19 Oct 2026 · Siemens, Erlangen

Factories in high-wage countries need 7–9 % more productivity every year.

Siemens invests €500 million in Erlangen as a showcase for AI in manufacturing.

Sources: Siemens press release, 21 Oct 2026; technology.org, 20 Oct 2026; Pfab et al. 2024, Data in Brief 53:110110

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PROJECT SCOPE

What we will and will not do

Aim

Build and test an AI decision gate that releases AOI false calls on a real Siemens line, while at most 1 % of defective boards are released, also when the line changes over time.

Includes

- Siemens AOI data (132 days, one line)
- Decision gate: trees vs. neural network
- Replay simulation of the line in time order
- Drift alarm and update policies
- Error analysis, replay demo

Out of scope (current phase)

- Changing the AOI machine or its camera images
- Other lines, plants or companies' own data
- Running the gate on a real line
- Root-cause analysis of solder defects

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GOALS

Measurable targets, fixed before the main experiments

≤ 1 %

Defective boards released

A released defect reaches the customer: the strictest limit. Baseline replay: 2.4–4.6 %.

≥ 25 %

False-call boards released

Each one is a manual inspection saved. Baseline (time order): 8.2 %.

Alarm

Drift detected

The gate warns when the line has changed and its decisions may no longer hold.

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OBJECTIVES

What the project will achieve

1. Build an AI decision gate that decides per flagged board: release or manual inspection.
2. Build a replay simulation that plays the Siemens line back day by day in time order.
3. Compare gradient-boosted trees with a neural network, and compare model update policies.
4. Add a drift alarm that warns when the line has changed.
5. Evaluate with production KPIs: defective boards released, manual inspections saved, time per board.

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KEY FACTORS

What decides whether the gate helps the line

False-call rate	Each false call costs operator time at the manual station
Missed defects	A released defective board reaches the customer: must stay ≤ 1 %
Drift over time	Materials, settings and products change; last month’s model may not fit
Class imbalance	Only about 1 % of flagged joints are real defects
Label quality	Labels come from operators and may contain errors

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INPUT → SYSTEM → OUTPUT

Where the gate sits on the line

On the line today

SMT soldering

AOI camera checks every joint against fixed limits

Flags suspicious boards

INPUT

AOI data per board

~70 measurements per joint

Inspection type

Component type

AOI error code

SYSTEM (OURS)

AI decision gate

Classifier trained on past manual-inspection results

Release threshold: at most 1 % of defective boards released

OUTPUT

Release

likely false call → next step

Manual inspection

possible defect → operator checks

Today

Every flagged board is checked by hand, and 98 % of them are fine.

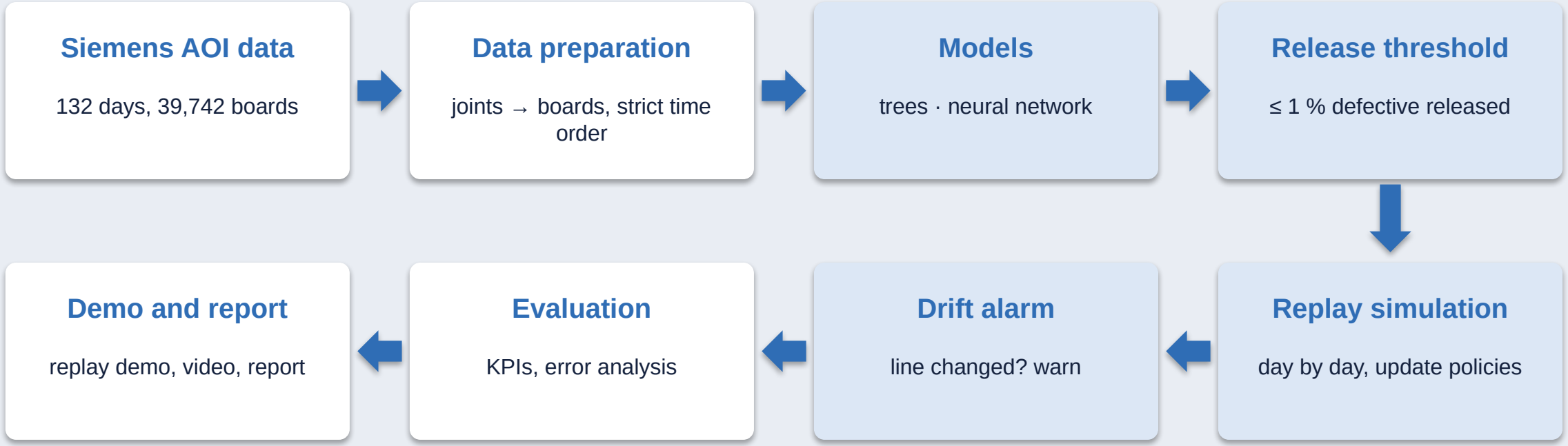
With the gate (goal)

Fewer manual checks, same quality: at most 1 % of defective boards released.

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PROJECT FLOW CHART

From the Siemens data to production KPIs



Blue boxes: the system we build. White boxes: data and project steps.

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METHODOLOGY

CRISP-DM, evaluated by a replay simulation of the line

1 · Business understanding	2 · Data understanding	3 · Data preparation	4 · Modelling	5 · Evaluation	6 · Deployment
false-call problem, KPIs, limits	structure, imbalance, drift	joints → boards, strict time order	gradient-boosted trees; neural network	replay simulation, production KPIs	replay demo, run instructions

Replay simulation (our “baseline simulation”)

The 132 recorded days are played back in time order. Each day the gate decides every flagged board with a model trained only on earlier days, exactly as on the real line.

We compare update policies (train once, retrain weekly on all past days, retrain on recent days) and measure released false calls, released defects and saved inspections.

Why not a random split? It mixes future and past and overstates what a factory gets (next slides).

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DATA

Real production data published by Siemens

Source	Pfab, Eichler, Mallandur, Rothering (2024), Siemens AG, Data in Brief 53, 110110
Line	One SMT production line of Siemens AG in Bavaria, 132 days
Size	440,274 flagged solder joints on 39,742 boards; 5 inspection types
Features	~70 AOI measurements + component type, error code, angle (anonymised, scaled 0–1)
Label	Manual inspection result: false call or true defect (875 defective boards)
Drift	Early vs. late days separable from measurements alone: AUROC 0.966
Licence	CC BY-NC 3.0 (Siemens AG): downloaded by script, not redistributed

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TECHNOLOGY STACK

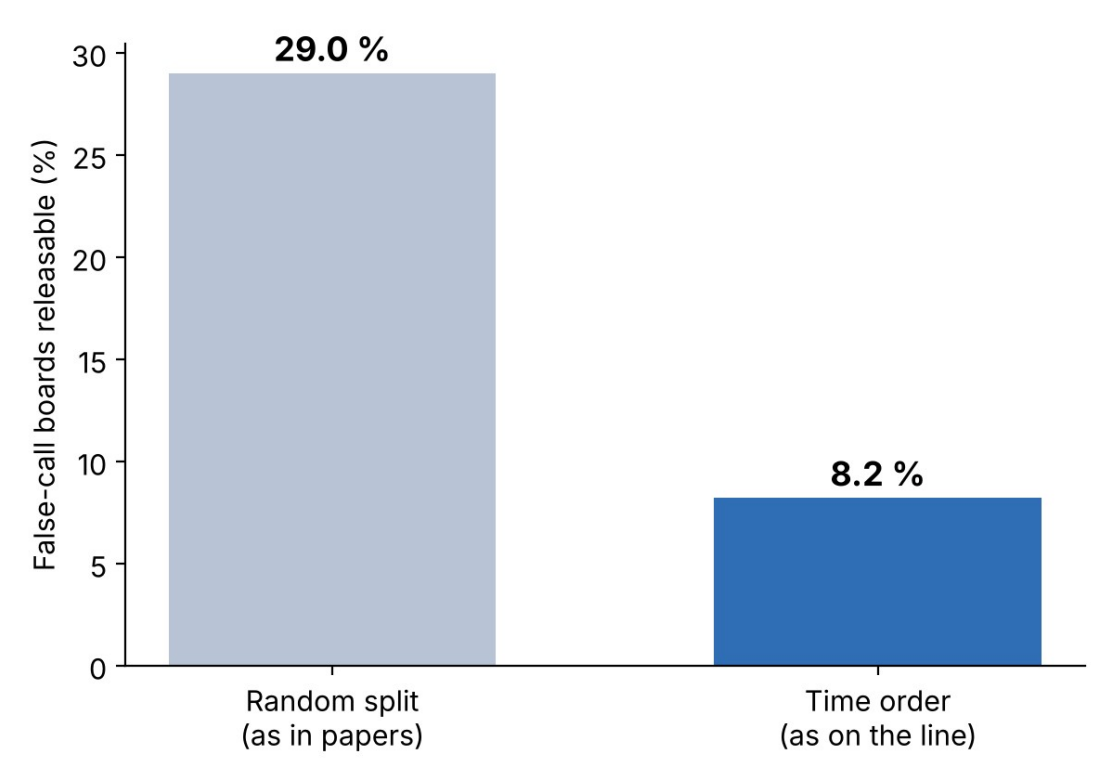
Open source, runs on a normal computer

Process	Tool	Purpose
Data	Python, pandas	Load the Siemens data, group joints into boards, keep time order
Models	scikit-learn, PyTorch	Gradient-boosted trees (baseline) and a neural network
Simulation	Python (own replay code)	Play the line back day by day; measure KPIs per policy
Drift alarm	Python, statistics	Detect when the line has changed
Quality	pytest, GitHub Actions	Tests run automatically on every change
Demo	Gradio or Streamlit	Replay the line and show decisions and saved inspections
Code sharing	GitHub	github.com/bodhi584/aoi-decision-gate

Within the lecture topics: neural networks (p. 12, 17), big data processing (p. 10). No large language model. Tools may change as the project progresses.

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Time order is much harder than a random split



What it shows

Gradient-boosted trees, trained on the first 70 % of records.

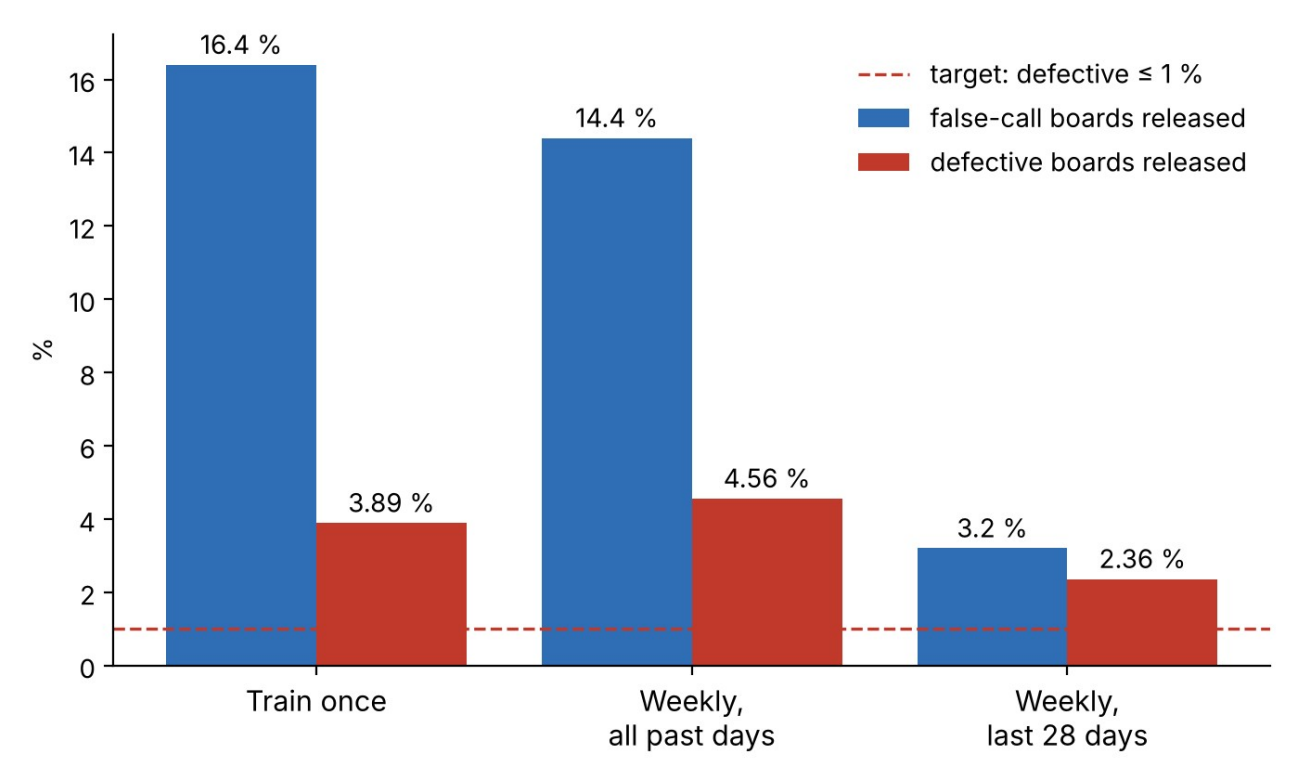
Share of false-call boards that could be released while 99 % of defective boards still go to manual inspection.

Random split: 29 %, AUROC 0.966. Time order: 8.2 %, AUROC 0.889.

The line changes over time: drift is the real challenge, and papers with random splits overstate results.

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Thresholds from past days do not hold on later days



What it shows

39 production days, 17,366 flagged boards replayed.

Up to 16 % of false-call boards released (2,750 manual inspections saved).

But 2.4–4.6 % of defective boards released: the 1 % limit is broken by drift.

→ Core of the project: drift detection and update policy.

WORKING METHODOLOGY

Four steps, each with a result we can show

Step 1 · Data preparation

Download the Siemens AOI data, check checksum
Group solder joints into boards; encode categories
Keep strict time order; facts on imbalance and drift (done)

Step 2 · Model development

Gradient-boosted trees as baseline (running)
Neural network in PyTorch
Release threshold: at most 1 % defective boards released

Step 3 · Simulation integration

Replay the line day by day, models trained on past days only
Compare update policies; add drift alarm
Count released false calls and released defects per day

Step 4 · Evaluation and comparison

Production KPIs per model and policy
Error analysis by inspection type, component, period
Replay demo, video and report

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TEAM

Who does what

Yu Han · team lead

Decision gate and replay simulation (core code), GitHub and tests

Melisa Subasi

Data understanding, evaluation and KPIs, final report editor

Meron Hagos

Literature review, industrial case (Siemens), demo interface

Alejandro Rivas

Experiments: models, drift handling and update policies

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TIMELINE

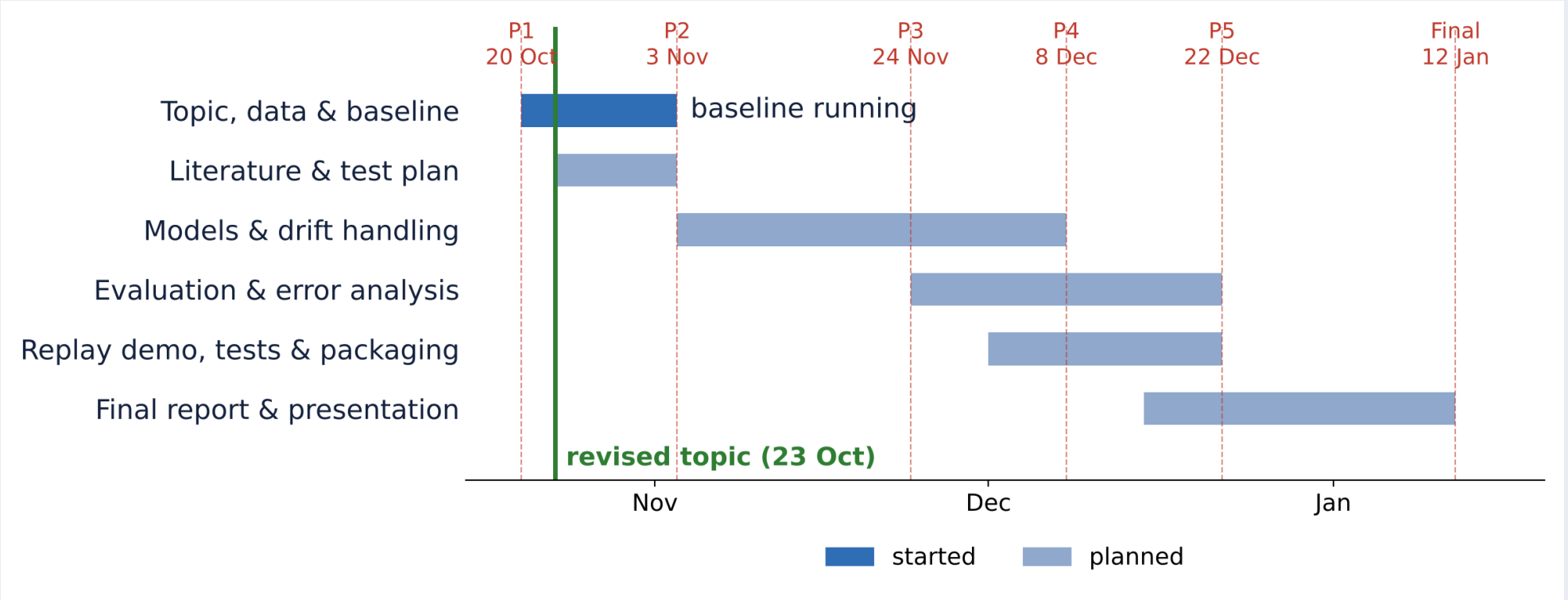
From proposal to final hand-in (lecture p. 33)

Date	Milestone	What we deliver
20 / 23 Oct	P1 · Proposal + tech stack + MVP	Revised topic, proposal, GitHub, baseline and first replay running
3 Nov	P2 · Core code + baseline sim; test plan + report	Decision gate and replay as core code, all update policies, test plan, literature report
24 Nov	P3 · Feature progress; unit/CI; issues board	Neural network, drift alarm, tests on every push
8 Dec	P4 · Feature progress; unit/CI; issues board	Full evaluation and error analysis
22 Dec	P5 · Demo video + run instructions; packaging	Replay demo, video, run instructions, package
12 Jan	Final presentation, report	Final presentation and report

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GANTT CHART

Project schedule



EXPECTED OUTCOMES

What we hand in, concretely: a software prototype

1 · Software: the AI decision gate

A Python package with a command-line tool. Train it on past inspection records; give it the AOI data of one board; it answers RELEASE or INSPECT, with a score and the main reasons.

2 · Replay dashboard (demo)

A small web app: pick a day of the Siemens line and see each board's decision, manual checks saved, defects released and drift alarms.

```
# planned command-line interface (day 87 = real replay result)

$ aoi-gate train   --data history.csv
  model saved, threshold set (≤ 1 % defects released)
$ aoi-gate decide  --board board_10466.csv
  RELEASE   (score below threshold: likely false call)
$ aoi-gate replay  --policy weekly
  day 87: 612 boards, 183 checks saved, 0 defects released
```

3 · Evidence and documents

Result files for every number · tests on every push · demo video · final report · public GitHub repository

In a factory it would run between the AOI machine and the manual inspection station. Connecting it to a real line is not part of the course project. The command names are planned; the day-87 numbers come from our first replay (weekly retraining).

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BILL OF MATERIALS (BOM)

What the project needs

Component	Specification	Quantity	Cost
Data set	Siemens AOI data, Mendeley (CC BY-NC 3.0), 333 MB	1	Free
Computer	Normal laptop or Google Colab CPU (no graphics card needed)	4 (one per member)	Free
Software	Python and open-source libraries	–	Free
Code hosting and tests	GitHub, GitHub Actions	1 repository	Free

Total cost: 0 €

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What the project covers, and what it does not

Conditions

One Siemens SMT line, five inspection types, 132 days
Gate placed after the AOI, before manual inspection
Decisions per board; at most 1 % defective boards released
Public data only, no personal data

Limitations

Measurements anonymised: no physical names
Labels from operators may contain errors
No images; no other lines or plants
Licence non-commercial (course use only)
Results hold for this line; other lines need their own data

SAFETY & ETHICAL CHECKLIST

Keeping it safe and realistic

Checklist

- ✓ Software and simulation only: no machinery, no physical risk
- ✓ Data anonymised by Siemens; no personal data
- ✓ Data used under its licence (non-commercial), not redistributed
- ✓ Safety first in the design: limit on released defects
- ✓ Transparent: every number from a result file; code public
- ✓ Reproducible: fixed seeds, tests on every push; all sources cited

Main risks and fallback

Drift limits how many false calls can be released safely → fallback: report which drift handling works, which does not, and why

Labels may contain operator errors → error analysis checks suspicious cases

In a real factory a gate needs approval by quality engineers before it may release boards

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FUTURE WORK

Next steps after the course

One gate per inspection type

The data has 5 inspection types with very different defect rates; a separate threshold or model per type may release more false calls safely.

Threshold from real costs

Set the threshold from the cost of one manual check vs. one escaped defect, and report savings in operator hours.

Suspicious labels

Find likely operator labelling errors and measure how much they limit the gate.

Shadow mode in a factory

The step before any real use: the gate runs next to the operators and only logs its decisions, so quality engineers can compare before it may release boards.

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Lecture slides, Case Study Intelligent Systems in Production, p. 10, 12, 17, 18, 33.

Thank you

github.com/bodhi584/aoi-decision-gate

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